CEGM2003 Case Study Description

Identifying Functional Port Areas Using AIS and Satellite Imagery data

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1 Introduction

Accurate identification of port areas and infrastructure, such as berths, jetties, breakwaters, anchorage zones, and basins, is essential for maritime safety, operational efficiency, and strategic port development. However, publicly available documentation of such facilities is often incomplete or outdated, motivating the use of data-driven approaches for their detection and localization [1].

Automatic Identification System (AIS) data and satellite imagery are two complementary sources that can help achieve this goal. AIS data capture real-time vessel positions and movements, revealing behavioral patterns such as slowing down, maneuvering, or stopping, which may indicate berthing or anchoring activities. These patterns can be analyzed using clustering techniques to infer the locations of functional port areas such as berths, anchorage zones, turning basins, or traffic channels.

Recent studies such as Hadjipieris et al. [1], have proposed unsupervised methods that cluster AIS data to identify berthing sites, showing promising results but also highlighting key challenges. Different port areas exhibit distinct spatial and behavioral signatures, meaning that a single clustering approach rarely generalizes across all cases. Moreover, the validation of clustering results requires additional datasets, such as labelled port areas or satellite imagery. Satellite imagery provides a complementary perspective: it captures static physical infrastructure that can be segmented or classified to delineate port features. By combining this with AIS-derived behavioral insights, one can assess how well movement-based patterns align with visible structures. Despite their potential, AIS and satellite data are rarely integrated or jointly analyzed in port studies.

2 Datasets and materials

The project uses two datasets: (1) AIS data from U.S. Coast Guard for the Port of Los Angeles area in August 2024, and (2) satellite imagery data from Sentinel-2 or PlanetScope.

2.1 AIS data

Dataset specifics: The AIS dataset includes static and dynamic vessel information. Static attributes comprise Maritime Mobile Service Identity (MMSI), vessel length, width, and type. Dynamic attributes include latitude, longitude, speed over ground (SOG), course over ground (COG), and heading.

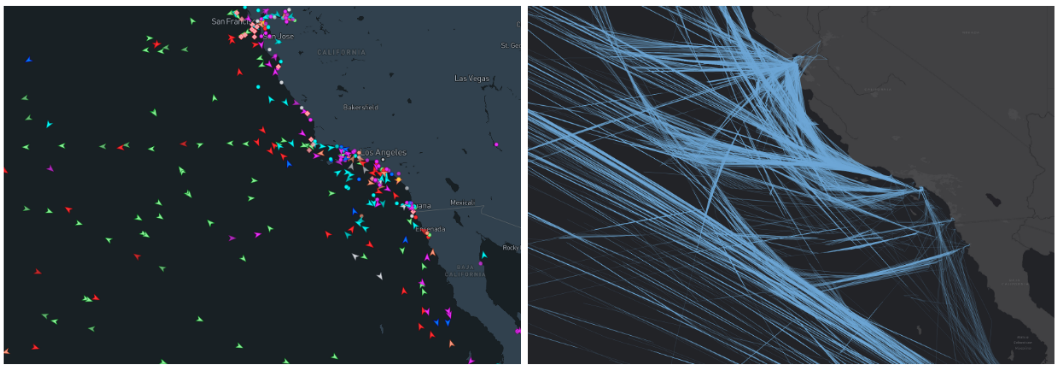

Figure 1: Visualization of AIS data along the U.S. West Coast (Left: snapshot of vessel positions. Right: vessel trajectories)

Pre-processing: The AIS data are spatially filtered to the physical area bounded by coordinates 33.60–33.85° N and 118.45–118.15° W, and temporally limited to 1st to 30th of August 2024. So, the students will receive a dataset that is filtered by area and time, with duplicates and anomalies removed. They are expected to interpolate and standardize the data according to the selected clustering method.

Analysis and unsupervised learning: Clustering AIS data can be performed using point-based [1] and trajectory-based [2, 3] methods. This project primarily focuses on point-based clustering, as it is generally simpler and faster, making it effective for identifying spatial density patterns and high-traffic areas, such as berthing or anchoring. Although point-based methods overlook temporal and directional context, they are well-suited for large datasets where the main objective is to understand spatial density. Trajectory-based clustering, in contrast, can capture vessel dynamics and behavioral patterns (e.g., turning, maneuvering, or route following) but may demand more extensive pre-processing and greater computational resources. Given the project's emphasis on interpretability and exploratory analysis, point-based clustering provides an effective and efficient starting point. Common algorithms include DBSCAN, HDBSCAN, spectral clustering, and Gaussian Mixture Models, though students are encouraged to explore alternative techniques as well.

2.2 Satellite data

Dataset specifics: Optical imagery from Sentinel-2 (10m) and PlanetScope (3m) is used to provide a direct visual representation of port infrastructures (e.g., berth, container yards, logistics areas). Revist frequency is not a critical factor for this project, but the main objective here is to obtain a clear visual of port environment.

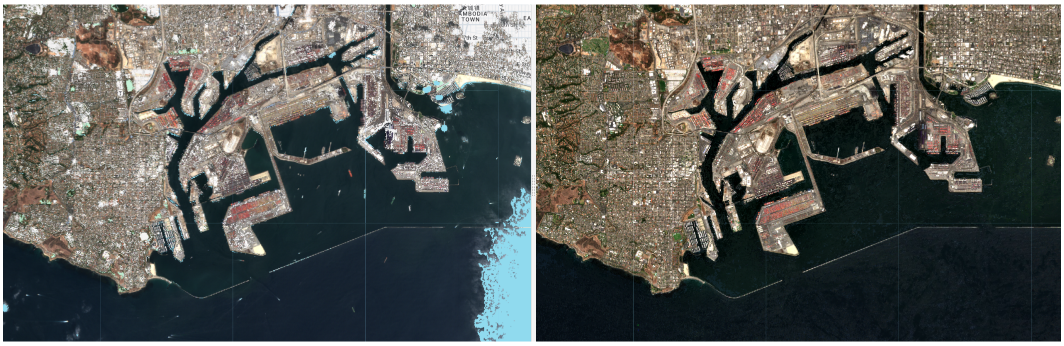

Figure 2: Pre-processed satellite data around port of LA (Left: cloud & shadow masking; Right: Mosaic imagery)

Pre-processing: Students apply light pre-processing such as cloud and shadow masking with creation of mosaics around the AOI. Data should also be standardized as needed, such that the zoom levels can support subsequent labeling and comparisons.

Cross-validation: Unlike AIS data, satellite imagery provides a direct visual representation of port infrastructure. Student will first delineate infrastructure areas using manual labeling and/or auto masking tools, then evaluate label quality. Once the masks meet a quality threshold, students can cross-reference whether the cluster shapefile of the AIS data falls within the dedicated port areas. This allows practical validation of clustering methods against the physical layout of the port. Quality of the results can be assessed by metrics such as Intersection-over-Union, Precision or Recall.

3 Learning objectives and potential research directions

The main goal of this project is to use AIS and satellite data to identify and localize functional port areas. The project is exploratory in nature and encourages students to experiment with different techniques for spatial clustering, pattern recognition, and validation. Through this work, students will develop technical and analytical skills in data science and AI applied to a real-world problem.

Learning Objectives

After completing the project, students should be able to have competency in the following areas:

1. Data Handling and Pre-processing: Clean, interpolate, and standardize large geospatial datasets (AIS trajectories and satellite images).
2. Modeling and Analysis: Applying and comparing multiple unsupervised clustering methods to identify patterns.
3. Validation and Evaluation: Using satellite-based segmentation to validate clustering outcomes by spatial metrics such as Intersection-over-Union, Precision, and Recall.
4. Critical Evaluation: Assessing the strengths and limitations of different methods and data sources for port area mapping.
5. Collaboration and Communication: Working effectively in a team to document workflows, interpret findings, and present conclusions.

Research Directions

The project is organized around three complementary tracks. The following guiding questions are provided to stimulate thinking and help structure the investigation at each stage.

AIS Data Analysis

- How can unsupervised learning be used to identify functional port areas from AIS patterns?
- Can clustering techniques distinguish between different port activities (e.g., berthing, anchoring) based on position-wise properties (e.g., location, speed, heading)?
- How does vessel type (e.g., cargo, tanker, tugboat) influence clustering patterns and the spatial distribution of detected port areas?

Satellite Imagery Exploration

- How accurately can automatic segmentation or masking tools identify port infrastructure in optical imagery?
- How do annotation strategies and spatial resolution affect labeling accuracy?
- What are the main sources of uncertainty when delineating port boundaries from imagery?

Data Fusion and Cross-Validation

- To what extent do AIS-derived functional areas overlap with satellite-detected physical infrastructure?
- Can integrating AIS and satellite imagery improve the completeness or reliability of port mapping compared to using either dataset alone?

4 Workflow

The project follows an integrated workflow that connects data pre-processing, unsupervised learning, spatial analysis, and validation. The trajectory of work is illustrated in Figure 3. The group will follow a similar sequence of stages while applying its own methodological choices within each stage.

The workflow begins with data preprocessing and exploration. AIS data are cleaned and filtered to remove errors and ensure spatial–temporal consistency, while satellite images undergo basic preparation such as cloud masking and mosaicking to align with the AIS coverage. Next, students apply **point-based clustering** methods to the AIS positions to detect dense regions corresponding to functional port areas (e.g., berths, anchorage zones). The resulting clusters are **converted into polygons** (e.g., convex hulls or alpha shapes) for geospatial comparison.

In parallel, satellite imagery is segmented to outline visible port infrastructure. Both outputs are then integrated by overlaying AIS-derived polygons on satellite masks to assess spatial correspondence through visual and quantitative evaluation (e.g., Intersection-over-Union, precision, recall). The project concludes with the synthesis of results and critical reflection on accuracy, uncertainty, and method suitability.

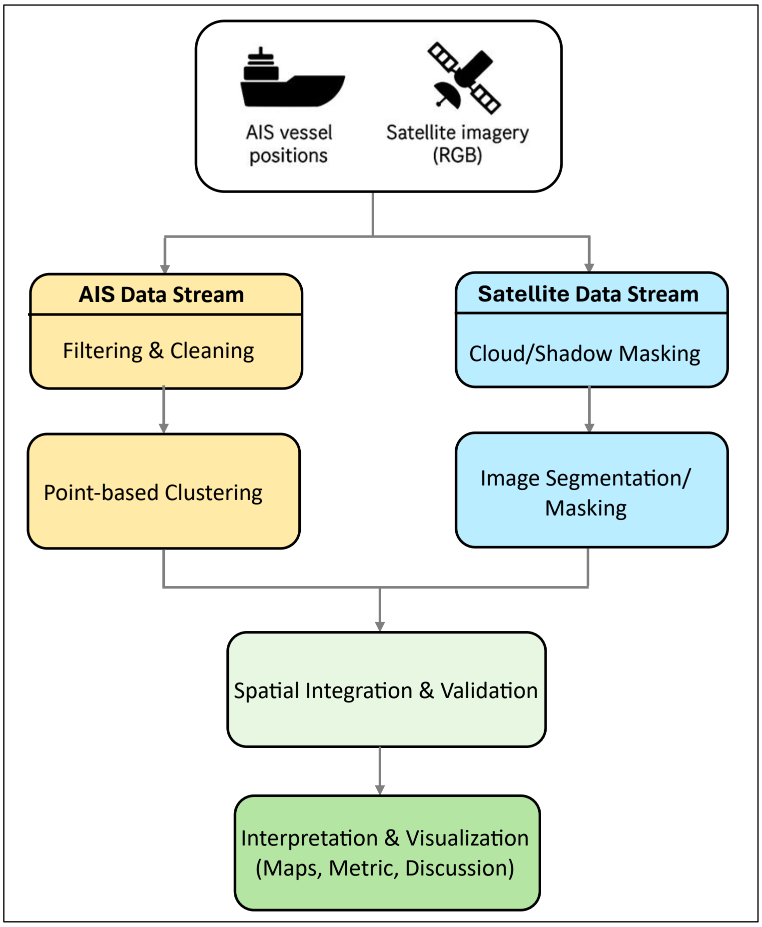

Figure 3: Conceptual workflow of the project

References

[1] Andreas Hadjipieris, Neofytos Dimitriou, and Ognjen Arandjelović. Unsupervised port berth identification from automatic identification system data, 2025. URL: https://arxiv.org/abs/2505.12046, arXiv:2505.12046.

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[3] Wayan Mahardhika Wijaya and Yasuhiro Nakamura. Port performance indicators construction based on the AIS-generated trajectory segmentation and classification. *International Journal of Data Science and Analytics*, 8 2024. doi:10.1007/s41060-024-00614-w.